

PROJECT REPORT

PLANNING, DESIGN AND DRAFTING OF A G+2 RESIDENTIAL BUILDING

PROJECT TITLE

PLANNING, DESIGN AND DRAFTING OF A G+2 RESIDENTIAL BUILDING

A Minor Project Report

Submitted in partial fulfilment of the requirements for the award of

Diploma in Civil Engineering

Submitted by

Sainik Bhaumick

Diploma in Civil Engineering

Under the Guidance of

Department of Civil Engineering

DECLARATION

I hereby declare that the project titled “Planning, Design and Drafting of a G+2 Residential Building” is an original piece of work carried out by me under the guidance of the Civil Engineering Department. The drawings, specifications, calculations, and planning concepts presented in this report are prepared solely for academic and portfolio purposes and have not been submitted elsewhere for any other qualification.

Name: Sainik Bhaumick

ACKNOWLEDGEMENT

I express my sincere gratitude to the faculty members of the Department of Civil Engineering for their continuous guidance and technical support throughout the completion of this minor project. I also thank my institute for providing the resources and learning environment necessary to complete this work successfully.

ABSTRACT

This project focuses on the planning, structural layout, and drafting of a G+2 reinforced concrete residential building with an emphasis on functionality, safety, and adherence to standard building practices. The project includes architectural planning, structural components such as isolated footing, column layout, stair detailing, and preparation of drawings using AutoCAD. The objective is to simulate real-world residential design standards that align with professional and site-level execution requirements.

1. INTRODUCTION

Residential buildings demand a balance between architectural planning, structural safety, cost efficiency, and user comfort. The aim of this project is to design a G+~~2~~⁴ residential building that satisfies fundamental civil engineering principles while maintaining practical usability.

This project reflects real drafting workflows followed in consultancy firms and construction organizations.

2. OBJECTIVES OF THE PROJECT

- To plan a functional G+~~2~~⁴ residential building
- To prepare architectural and structural drawings using AutoCAD
- To understand load transfer and foundation planning
- To develop practical drafting skills for professional application
- To follow standard building construction practices

3. SCOPE OF THE PROJECT

The scope of this project includes:

- Site planning
- Ground floor and first floor planning
- Roof planning
- Front elevation and sectional elevation
- Staircase detailing
- Isolated column footing detailing
- Door and window schedule
- Specification notes

Structural design calculations are limited to conceptual understanding and standard detailing.

4. SITE DETAILS

- Type of building: Residential
- Number of floors: Ground + 2¹
- Orientation and access: As per planning layout
- Road width: 7 m (as shown in site plan)

5. BUILDING PLANNING

5.1 Ground Floor Planning

The ground floor consists of:

- Living area
- Bedrooms
- Kitchen
- Toilet and bathroom
- Staircase

The layout ensures proper ventilation, circulation space, and privacy.

5.2 First Floor Planning

The first-floor mirrors residential functionality with bedrooms and associated facilities accessed through the internal staircase.

5.3 Roof Planning

The roof plan includes:

- Stair headroom
- Water tank area
- Proper slope provision for drainage

6. STRUCTURAL COMPONENTS

6.1 Foundation

Isolated column footing is adopted to safely transfer loads from superstructure to subsoil.

Footing details include:

- Column size
- Footing dimensions
- Reinforcement detailing

6.2 Columns

RCC columns are placed symmetrically to ensure balanced load distribution and structural stability.

6.3 Staircase

A dog-legged staircase is planned, providing comfortable rise and tread as per standards.

7. DRAWING DETAILS

The following drawings are prepared using AutoCAD:

- Site plan
- Ground floor planning
- First floor planning
- Roof plan
- Front elevation
- Sectional elevation
- Isolated column footing detail
- Staircase detail
- Door and window schedule

All drawings are prepared to scale and follow standard drafting conventions

.

8. SPECIFICATIONS (BRIEF)

- Cement: OPC / PPC of approved quality
- Concrete: Nominal mix as per standard practice
- Steel reinforcement: HYSD bars
- Brickwork: First class bricks in cement mortar
- Plaster: Cement sand plaster
- Flooring: Vitrified / ceramic tiles
- Doors and windows: As per schedule

9. TOOLS & SOFTWARE USED

- AutoCAD (for drafting and detailing)
- Standard IS codes (for conceptual reference)

10. LEARNING OUTCOMES

- Practical exposure to residential building planning
- Improved drafting and drawing presentation skills
- Understanding of real-world construction detailing
- Portfolio-ready professional documentation

11. CONCLUSION

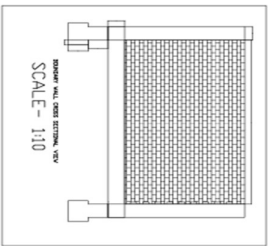
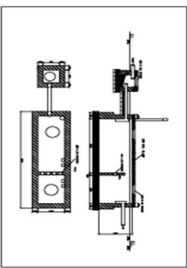
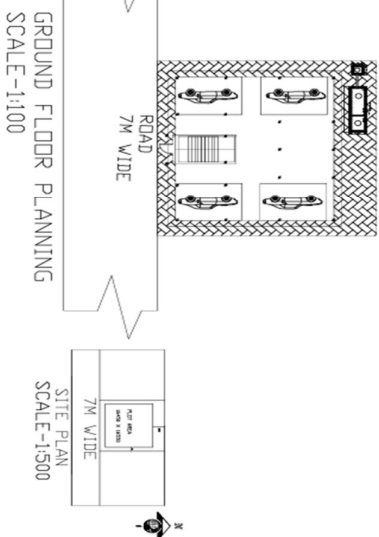
This project successfully demonstrates the planning and drafting of a G+21 residential building using professional standards. The drawings and layout reflect real industry practices and provide a strong foundation for future work in design offices and construction sites. This project enhanced both technical understanding and drafting proficiency.

— END OF PROJECT REPORT —

Prepared by:

Sainik Bhaumick

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Door & Window Schedule					
	WIDTH	HEIGHT	SILL	DESCRIPTION	
	1500	2100	N/A	MAIN ENTRANCE DOOR	
	1000	2100	N/A	BEDROOM ADJACENT DOOR	
	850	2100	N/A	KITCHEN DOOR TOILET DOOR	
	1900	1250	800	5	
	1300	1250	800	BEDROOM WINDOW	
	850	600	1300	KITCHEN & TOILET WINDOW	
	900	1350	800	BEDROOM & KITCHEN WINDOW	
SPECIFICATION:-					
	• 250 MM THICK DUTY BRICK				
	• 150MM THICK INNER BRICK				
	• WALL				
	• GRADE OF CONCRETE M20				
	• GRADE OF STEEL F2-45				
NOTE:-					
	• ALL DIMENSION IN MM				
	• ALL DUTY WALL 250 THICK				
	• ALL PARTITION WALL ARE 150 MM THICK				
	• ALL SKEPS IS TO BE TAKEN DURING CONSTRUCTION				
(Drawn C.I and Type 45 Separator)					
	Sheet No. _____ Project Name _____ Client Name _____ Date _____ Scale _____ Prepared By _____ Checked By _____ Approved By _____ Signature Area _____ _____ _____ _____ _____ _____ _____ _____				
A/C GRP	TITLE MINOR PROJECT				
IN TIME/LATE					
CDWL	DESIGNED BY: GROUP C				
CDMP	YEAR AND BRANCH DEC				
CHECKED BY					